

History

SQLite is the easiest database for beginners to learn. It is a powerful relational database management system (RDBMS) with a light and easy design. It is also the simplest database, that is perfect for practicing joins and simple queries.

The 1960s – beginnings, but the history of databases as we know them, really begins in **1970**

Charles Bachman designed the first computerized database in the early 1960s. This first database was known as **the Integrated Data Store**, or IDS. This was shortly followed by the Information Management System, a database created by IBM.

The first open source database management system was **MySQL** in 1995.

Unit I

Introduction

Characteristics of Database approach

Actors on the Scene

Workers behind the scene

Advantages of using DBMS approach

Data models

schemas and instances

Three-schema architecture and data independence

Database languages and interfaces

The database system environment

Centralized and client-server architectures

Classification of Database Management systems,

Entity-Relationship Model:

Using high level Conceptual data models for Database Design

A Sample Database Application

Entity types, Entity sets Attributes and Keys

Relationship types, Relationship Sets,

Roles and Structural Constraints

Weak Entity Types.

INTRODUCTION (1)

- Database - collection of related data (known facts that can be recorded and that have implicit meaning)
- A database has the following implicit properties:
 - represents some aspect of the real world, called the miniworld or the universe of discourse (DoD). Changes to the miniworld are reflected in the database.
 - logically coherent collection of data with some inherent meaning.
 - designed, built, and populated with data for a specific purpose. It has an intended group of users and applications
- A database can be of any size and of varying complexity – address book, library, tax payers information

INTRODUCTION (2)

- Traditional databases applications – information that is stored and accessed is either textual or numeric ex:- Bank, hotel or airline reservation, library, book, toy, online shopping, supermarket.
- Multimedia databases - store pictures, video clips, and sound messages.
- Geographic information systems (GIS) can store and analyze maps, weather data, and satellite images.
- Data warehouses and online analytical processing (OLAP) systems - decision making.
- Real-time and active database technology - controlling industrial and manufacturing processes.
- World Wide Web - search for information.
- Used in business, electronic commerce, engineering, medicine, law, education, and library science.

Typical DBMS Functionality

Define a particular database in terms of its data types, structures, and constraints

Construct or Load the initial database contents on a secondary storage medium

Manipulating the database:

- Retrieval: Querying, generating reports
- Modification: Insertions, deletions and updates to its content
- Accessing the database through Web applications

Processing and *Sharing* by a set of concurrent users and application programs – yet, keeping all data valid and consistent

INTRODUCTION (3)

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- A database management system (DBMS) is a collection of programs that enables users to create and maintain a database.
 - Facilitates the processes of defining, constructing, manipulating, and sharing databases among various users and applications.
 - Defining - specifying the data types, structures, and constraints
 - Constructing - storing the data itself on some storage medium
 - Manipulating - querying, updating, and generating reports from the data.
 - Sharing - multiple users and programs to access the database concurrently.
 - Protection - hardware or software malfunction and security protection against unauthorized or malicious access.
 - Maintain - evolve as requirements change over time.
 - DBMS software - General-purpose or special-purpose.

INTRODUCTION (4)

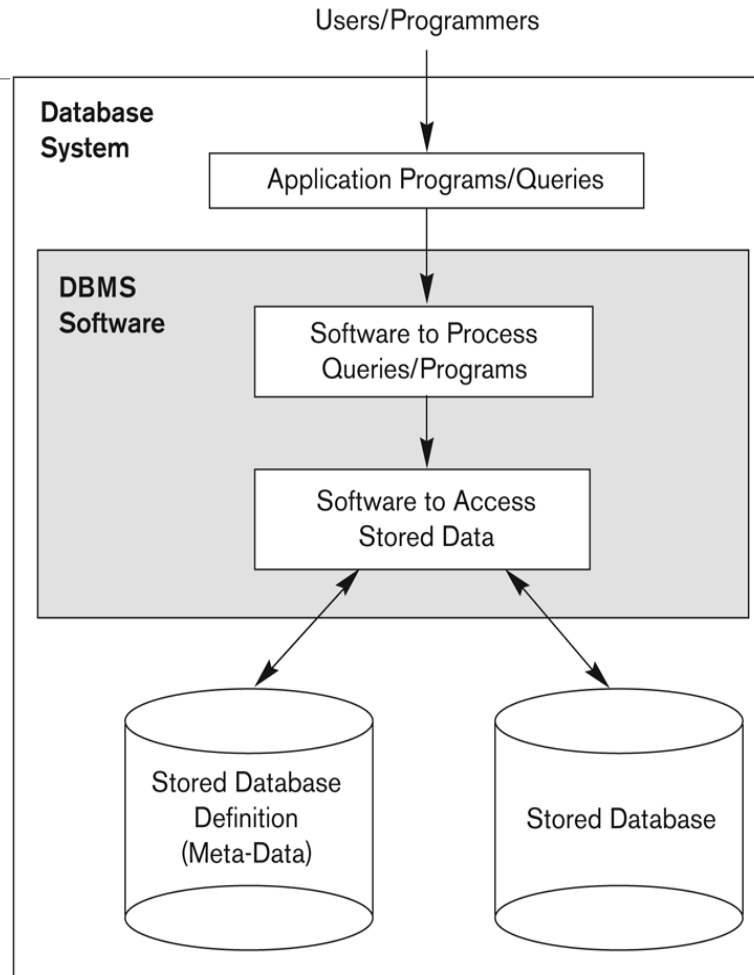


Figure 1.1
A simplified database
system environment.

INTRODUCTION (5)

UNIVERSITY database for maintaining information concerning students, courses, and grades in a university environment.

STUDENT - stores data on each student

COURSE - stores data on each course

SECTION file - stores data on each section of a course,

GRADE_REPORT - stores the grades

PREREQUISITE - stores the prerequisites of each course.

INTRODUCTION (6)

STUDENT

Name	Student_number	Class	Major
Smith	17	1	CS
Brown	8	2	CS

COURSE

Course_name	Course_number	Credit_hours	Department
Intro to Computer Science	CS1310	4	CS
Data Structures	CS3320	4	CS
Discrete Mathematics	MATH2410	3	MATH
Database	CS3380	3	CS

SECTION

Section_identifier	Course_number	Semester	Year	Instructor
85	MATH2410	Fall	07	King
92	CS1310	Fall	07	Anderson
102	CS3320	Spring	08	Knuth
112	MATH2410	Fall	08	Chang
119	CS1310	Fall	08	Anderson
135	CS3380	Fall	08	Stone

GRADE_REPORT

Student_number	Section_identifier	Grade
17	112	B
17	119	C
8	85	A
8	92	A
8	102	B
8	135	A

PREREQUISITE

Course_number	Prerequisite_number
CS3380	CS3320
CS3380	MATH2410
CS3320	CS1310

Figure 1.2
 A database that stores
 student and course
 information.

CHARACTERISTICS OF DATABASE APPROACH (1)

Student details maintained by faculty, account section and dept. office may have different attributes.

Having multiple copies using traditional approach will lead to redundancy and inconsistency.

Characteristics of Database approach are:

- Self-describing nature of a database system
- Insulation between programs and data, and data abstraction
- Support of multiple views of the data
- Sharing of data and multiuser transaction processing

CHARACTERISTICS OF DATABASE APPROACH (2)

Self Describing Nature of a Database System

- Complete definition or description of the database structure and constraints – **metadata**
- **DBMS catalog** stores metadata
- Catalog is used by the DBMS software and database users
- In traditional file processing, structure is declared in the application programs.

RELATIONS

Relation_name	No_of_columns
STUDENT	4
COURSE	4
SECTION	5
GRADE_REPORT	3
PREREQUISITE	2

COLUMNS

Column_name	Data_type	Belongs_to_relation
Name	Character (30)	STUDENT
Student_number	Character (4)	STUDENT
Class	Integer (1)	STUDENT
Major	Major_type	STUDENT
Course_name	Character (10)	COURSE
Course_number	XXXXNNNN	COURSE
....		
....		
....		
Prerequisite_number	XXXXNNNN	PREREQUISITE

Note: Major_type is defined as an enumerated type with all known majors. XXXXNNNN is used to define a type with four alpha characters followed by four digits

Figure 1.3

An example of a database catalog for the database in Figure 1.2.

CHARACTERISTICS OF DATABASE APPROACH (3)

Insulation between Programs and Data, and Data Abstraction

- **Program-data independence** – no need to change program if the structure of the file changes. For example – adding DoB to student data.
- An **operation** (also called a function or method) is specified in two parts.
- The **interface** of an operation includes the operation name and the data types of its arguments.
- The **implementation** of the operation is specified separately.
- **Program-operation independence** – changing the implementation of the operation without affecting its interface. For ex – calculating grade of student.
- **Data abstraction** - The characteristic that allows program-data independence and program-operation independence.
- **Data model** – is used to provide data abstraction to database users, it provides conceptual view without disclosing storage details.

CHARACTERISTICS OF DATABASE APPROACH (4)

Support of Multiple Views of the Data

- Many users - require a different perspective or view of the database.
- A view may be a subset of the database or it may contain virtual data that is derived from the database files but is not explicitly stored.
- Some users may not need to be aware of whether the data they refer to is stored or derived.

CHARACTERISTICS OF DATABASE APPROACH (5)

Sharing of Data and Multiuser Transaction Processing

- A multiuser DBMS -allows multiple users to access the database at the same time.
- Concurrency control software – Controlled manner updation of database
- Example – Bus ticket reservation
- OLTP – Online Transaction Processing
- Transaction - executing program or process that includes one or more database accesses, such as reading or updating of database records.
- Transaction properties –Atomicity, Consistency, Isolation and Durability

ACTORS ON THE SCENE (1)

Database Administrators

- Organisation's primary resource – database, secondary resource - DBMS and related software.
- Administering these resources is the responsibility of the database administrator (DBA).
- The DBA -
 - authorizing access to the database,
 - coordinating and monitoring its use,
 - acquiring software and hardware resources as needed.
 - breach of security
 - poor system response time.
- In large organizations, the DBA is assisted by a staff that helps carry out these functions

ACTORS ON THE SCENE (2)

Database Designers

- Identify the data to be stored in the database
- Choose appropriate structures to represent and store the data.
- Interact with each potential group of users
- Develop views of the database that meet the data and processing requirements of these groups.
- Each view is then analyzed and integrated with the views of other user groups.
- The final database design must be capable of supporting the requirements of all user groups
- Designing is mostly done before the database is implemented and populated with data.
- In many cases, they are on the staff of the DBA and may be assigned other staff responsibilities later.

ACTORS ON THE SCENE (3)

END USERS (1)

End users are the people whose jobs require access to the database for querying, updating, and generating reports. Categories of end users:

Casual end users

- occasionally access the database
- may need different information each time.
- Use a sophisticated database query language to specify their requests
- middle- or high-level managers or other occasional browsers.
- learn only a few facilities that they may use repeatedly

ACTORS ON THE SCENE (4)

END USERS (2)

Naive or parametric end users -

- Constant querying and updating the database,
- Using standard types of queries and updates-called canned transactions-that have been carefully programmed and tested.
- Bank tellers - check account balances and post withdrawals and deposits.
- Reservation clerks for airlines, hotels, and car rental companies - check availability and make reservations.
- Learn very little about the facilities provided by the DBMS
- Understand only the user interfaces of the standard transactions designed and implemented for their use.

ACTORS ON THE SCENE (5)

END USERS (3)

Sophisticated end users

- Include engineers, scientists, business analysts, and others
- Thoroughly familiarize themselves with the facilities of the DBMS to implement their applications
- Learn most of the DBMS facilities in order to achieve their complex requirements

ACTORS ON THE SCENE (6)

END USERS (4)

Stand-alone users

- Maintain personal databases by using ready-made program packages that provide easy-to-use menu-based or graphics-based interfaces.
- Example - user of a tax package - stores a variety of personal financial data for tax purposes.
- Usually become very proficient in using a specific software package.

ACTORS ON THE SCENE (7)

System Analysts and Application Programmers (Software Engineers)

- Determine the **requirements of end users**, especially naive and parametric end users
- Develop specifications for **canned transactions** that meet these requirements.
- Application programmers implement these specifications as programs; then they **test, debug, document, and maintain** these canned transactions.
- Such analysts and programmers-commonly referred to as software engineers-should be familiar with the **full range of capabilities** provided by the DBMS to accomplish their tasks

WORKERS BEHIND THE SCENE (1)

DBMS system designers and implementers

Design and implement the DBMS modules and interfaces as a software package. Responsible for:

- implementing the catalog,
- processing query language, processing the interface,
- accessing and buffering data,
- controlling concurrency,
- handling data recovery and security.

The DBMS must interface with other system software, such as the operating system and compilers for various programming languages.

WORKERS BEHIND THE SCENE (2)

Tool Developers

- Design and implement tools-the software packages
- Tools are optional packages that are often purchased separately.
- They include packages for database design, performance monitoring, natural language or graphical
- interfaces, prototyping, simulation, and test data generation.
- In many cases, independent software vendors develop and market these tools.

WORKERS BEHIND THE SCENE (3)

Operators and maintenance personnel

- System administration personnel who are responsible for the actual running and maintenance of the hardware and software environment for the database system.

Workers behind the scene are instrumental in making the database system available to end users, they typically do not use the database for their own purposes.

ADVANTAGES OF USING THE DBMS APPROACH (1)

Controlling Redundancy

In traditional software development utilizing file processing, every user group maintains its own files for handling its data-processing applications.

This **redundancy** in storing the same data multiple times leads to several problems.

First, entering data on a new student-multiple times: once for each file where student data is recorded. This leads to *duplication of effort*.

Second, *storage space is wasted* when the same data is stored repeatedly, and this problem may be serious for large databases.

Third, files that represent the same data may become *inconsistent*.

It is sometimes necessary to use **controlled redundancy** for improving the performance of queries

ADVANTAGES OF USING THE DBMS APPROACH (1)

Controlling Redundancy

Figure 1.6

Redundant storage of Student_name and Course_name in GRADE_REPORT.
(a) Consistent data.
(b) Inconsistent record.

GRADE_REPORT

Student_number	Student_name	Section_identifier	Course_number	Grade
17	Smith	112	MATH2410	B
17	Smith	119	CS1310	C
8	Brown	85	MATH2410	A
8	Brown	92	CS1310	A
8	Brown	102	CS3320	B
8	Brown	135	CS3380	A

(a)

GRADE_REPORT

Student_number	Student_name	Section_identifier	Course_number	Grade
17	Brown	112	MATH2410	B

(b)

ADVANTAGES OF USING THE DBMS APPROACH (1)

Restricting Unauthorized Access

- When multiple users share a large database, it is likely that most users will not be authorized to access all information in the database. Example –Financial data.
- some users may be permitted only to retrieve data, whereas others are allowed both to retrieve and to update
- Typically, users or user groups are given account numbers protected by passwords, which they can use to gain access to the database.
- A DBMS should provide a security and authorization subsystem, which the DBA uses to create accounts and to specify account restrictions.
- The DBMS should then enforce these restrictions automatically.
- only the DBA's staff may be allowed to use certain privileged software, such as the software for creating new accounts.
- Similarly, parametric users may be allowed to access the database only through the canned transactions developed for their use.

ADVANTAGES OF USING THE DBMS APPROACH (1)

Providing Persistent Storage for Program Objects

- Programming languages typically have complex data structures, such as class definitions in c++ or Java.
- The values of program variables are discarded or explicitly stored into a format suitable for file storage.
- To read this data once more, the programmer must convert from the file format to the program variable structure.
- Object-oriented database systems are compatible with programming languages such as c++ and Java, and the DBMS software automatically performs any necessary conversions.
- A complex object in c++ can be stored permanently in an OODB. Such an object is said to be persistent, since it survives the termination of program execution and can later be directly retrieved by another c++ program.
- The persistent storage of program objects and data structures is an important function of database systems.
- Traditional database systems suffered from impedance mismatch problem, OODB offers data structure compatibility with one or more object oriented programming languages.

ADVANTAGES OF USING THE DBMS APPROACH (1)

Providing Storage Structures for Efficient Query Processing

- DBMS must provide specialized data structures to speed up disk search for the desired records.
- Auxiliary files called indexes are used for this purpose.
- Indexes are typically based on tree data structures or hash data structures, suitably modified for disk search.
- In order to process the database records needed by a particular query, those records must be copied from disk to memory.
- DBMS has a buffering module that maintains parts of the database in main memory buffers or use operating system for the same.
- The query processing and optimization module of the DBMS is responsible for choosing an efficient query execution plan for each query based on the existing storage structures.
- Choice of index creation is part of physical database design and tuning, done by DBA staff.

ADVANTAGES OF USING THE DBMS APPROACH (1)

Providing Backup and Recovery

- A DBMS must provide facilities for recovering from hardware or software failures.
- The **backup and recovery subsystem** of the DBMS is responsible for recovery.
- For example, if the computer system fails in the middle of a complex update transaction,
- it should be recovered to initial state or resume from the point it was interrupted.